GATE ME 90 Days Challenge

Day 18/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	**Week 3 Revision**	**91 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on all five topics of Week 3: columns and energy methods; mechanism analysis; calculus; dynamics of linkages and flywheels; cams, gears and gear trains (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A solid steel column of 80 mm diameter and 4 m length has both ends fixed. For E = 210 GPa, the Euler buckling load (in kN, round off to one decimal place) is _____

Q2 MCQ

A simply supported beam of span L and flexural rigidity EI carries a central point load P. The strain energy stored in bending is

(A) $P^2L^3/(48EI)$ **(B)** $P^2L^3/(6EI)$

(C) $P^2L^3/(192EI)$ **(D)** $P^2L^3/(96EI)$

Q3 NAT

A weight of 100 N falls through 50 mm onto a collar at the lower end of a vertical steel bar 2 m long with a cross-sectional area of 1000 mm^2. For E = 200 GPa, the maximum stress induced in the bar (in MPa, round off to two decimal places) is _____

Q4 MCQ

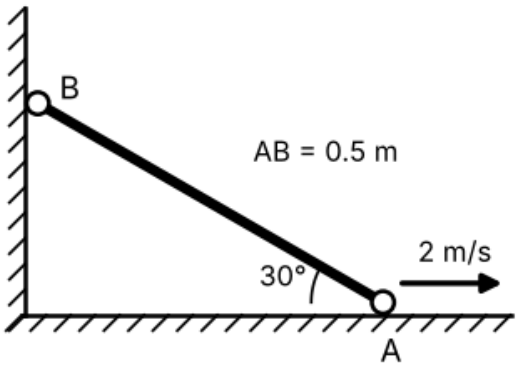
Fixing the slider (piston) of a single slider-crank chain gives

(A) the Whitworth quick-return mechanism **(B)** the pendulum pump (bull engine)

(C) the oscillating cylinder engine **(D)** the rotary (Gnome) engine

Q5 NAT

The rod AB shown (0.5 m long) slides with end A on the floor and end B against a vertical wall. When the rod makes 30° with the floor, end A moves away from the wall at 2 m/s. The speed of end B (in m/s, round off to two decimal places) is _____



Q6 MCQ

In a crank and slotted-lever quick-return mechanism, the distance between the fixed centres is 200 mm and the crank is 100 mm long. The ratio of the time of the cutting stroke to the time of the return stroke is

(A) 2.0
(B) 1.5
(C) 1.67
(D) 2.5

Q7 MCQ

The value of $\lim_{x\to 0} (\sin x / x)^{1/x^2}$ is

(A) 1
(B) $e^{1/6}$
(C) $e^{-1/6}$
(D) $e^{-1/3}$

Q8 NAT

The value of $\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$ (round off to three decimal places) is _____

Q9 MCQ

The area enclosed between the curves $y = x^2$ and $y = 2x$ is

(A) 8/3
(B) 2/3
(C) 4/3
(D) 4

Q10 NAT

During a punching operation a flywheel must give out 4.5 kJ of energy while its speed falls from 240 rpm to 216 rpm. If the radius of gyration of the flywheel is 0.6 m, its mass (in kg, round off to one decimal place) is _____

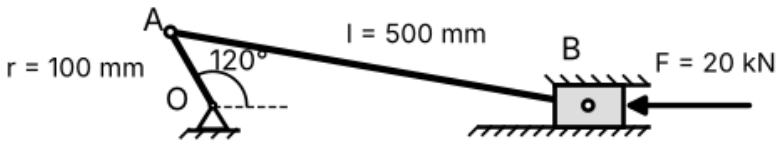
Q11 MCQ

A connecting rod is replaced by a dynamically equivalent system of two point masses at distances l_1 and l_2 on either side of its centre of gravity. If k is its radius of gyration about the centre of gravity, the condition to be satisfied is

(A) $l_1 l_2 = k^2$
(B) $l_1 + l_2 = k$
(C) $l_1 = l_2 = k/2$
(D) $l_1^2 + l_2^2 = k^2$

Q12 NAT

In the horizontal engine shown, the net force on the piston is 20 kN, the crank radius is 100 mm and the connecting rod is 500 mm long. With the crank at 120° from the inner dead centre, the turning moment on the crankshaft (in kN·m, round off to two decimal places) is _____



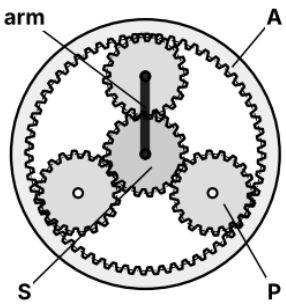
Q13 MCQ

A compound gear train has two stages: a 20-tooth pinion drives a 60-tooth gear, and an 18-tooth pinion on the same shaft as that gear drives a 54-tooth gear. If the input runs at 1440 rpm, the output speed is

(A) 480 rpm
(B) 240 rpm
(C) 120 rpm
(D) 160 rpm

Q14 NAT

In the epicyclic train shown, the sun S (20 teeth) is driven at 1000 rpm clockwise and the annulus A (60 teeth) at 200 rpm counter-clockwise. The speed of the arm carrying the planets (in rpm) is _____



S = sun (20 T) P = planets (20 T)
A = annulus (60 T, internal)

Q15 MCQ

If the speed of a cam is doubled while the lift, the cam angles and the follower motion stay the same, the maximum acceleration of the follower becomes

(A) 2 times
(B) 4 times
(C) the same
(D) 8 times

Answer Key

Q	Type	Answer
Q1	NAT	1041.8 (accept 1036 to 1047)
Q2	MCQ	(D) $P^2L^3/(96EI)$
Q3	NAT	31.72 (accept 31.5 to 31.9)
Q4	MCQ	(B) the pendulum pump (bull engine)
Q5	NAT	3.46 (accept 3.43 to 3.49)
Q6	MCQ	(A) 2.0
Q7	MCQ	(C) $e^{-1/6}$
Q8	NAT	0.098 (accept 0.097 to 0.099)
Q9	MCQ	(C) 4/3
Q10	NAT	208.3 (accept 206.3 to 210.3)
Q11	MCQ	(A) $I_1\, I_2 = k^2$
Q12	NAT	1.56 (accept 1.54 to 1.57)
Q13	MCQ	(D) 160 rpm
Q14	NAT	100 (clockwise)
Q15	MCQ	(B) 4 times

Detailed Solutions

Q1 **Answer: 1041.8 (accept 1036 to 1047)**

Concept: $P_{cr} = \pi^2 EI/L_e^2$, with $L_e = L/2$ for both ends fixed.

$I = \pi d^4/64 = \pi \times 80^4/64 =$ **2.0106×10^6 mm⁴**; $L_e = 4000/2 =$ **2000 mm**

$P_{cr} = \pi^2 \times 210\,000 \times 2.0106 \times 10^6 / 2000^2 = 1041811$ N = **1041.8 kN**

Exam tip: Effective lengths: both pinned L, fixed-free 2L, fixed-pinned L/√2, both fixed L/2. A fixed-fixed column is 4 times as strong as a pinned one of the same length.

Q2 **Answer: (D) $P^2L^3/(96EI)$**

Concept: $U = \int M^2/(2EI)\, dx$. By symmetry take twice the left half, where $M = Px/2$.

$U = 2 \int_0^{L/2} (Px/2)^2/(2EI)\, dx = (P^2/4EI) \times (L/2)^3/3 =$ **$P^2L^3/(96EI)$**

Check with Castigliano: $\delta = \partial U/\partial P = 2PL^3/(96EI) = PL^3/(48EI)$ ✓, the familiar central deflection.

Exam tip: For a single load, $U = P\delta/2$, so knowing $\delta = PL^3/48EI$ gives U directly. $P^2L^3/(6EI)$ is the cantilever result.

Q3 **Answer: 31.72 (accept 31.5 to 31.9)**

Concept: Impact loading: equate the work done by the falling weight to the strain energy: $W(h + \delta) = \sigma^2 AL/(2E)$, which gives $\sigma = (W/A)[1 + \sqrt{(1 + 2hAE/(WL))}]$.

$W/A = 100/1000 =$ **0.1 MPa**; $2hAE/(WL) = 2 \times 50 \times 1000 \times 200\,000/(100 \times 2000) =$ **100 000**

$\sigma = 0.1 \times (1 + \sqrt{100\,001}) = 0.1 \times 317.23 =$ **31.72 MPa**

Exam tip: Even a small drop multiplies the static stress (0.1 MPa) by over 300. With h = 0 (sudden load) the bracket becomes 2: a suddenly applied load doubles the stress.

Q4 **Answer: (B) the pendulum pump (bull engine)**

Concept: The four inversions of the single slider-crank chain: frame fixed → reciprocating engine; crank fixed → Whitworth quick-return, rotary engine; connecting rod fixed → oscillating cylinder engine, crank and slotted-lever quick-return; slider fixed → pendulum pump (hand pump).

So fixing the slider gives the **pendulum pump**.

Exam tip: Learn the list as 'frame, crank, rod, slider' and their two examples each. GATE often asks the odd one out.

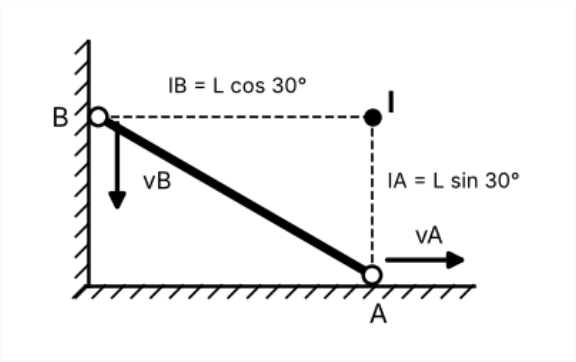
Q5 **Answer: 3.46 (accept 3.43 to 3.49)**

Concept: The instantaneous centre I lies where the normals to the two velocities meet: the vertical through A and the horizontal through B (figure).

$IA = L \sin 30° = 0.25$ m → $\omega = v_A/IA = 2/0.25 =$ **8 rad/s**

$IB = L \cos 30° = 0.433$ m → $v_B = \omega \times IB = 8 \times 0.433 =$ **3.46 m/s** (downward)

Exam tip: Shortcut: $v_B/v_A = \cot\theta = \cot 30° = 1.732$, where θ is the angle with the floor.



Q6 **Answer: (A) 2.0**

Concept: At the extreme positions the crank is perpendicular to the slotted lever, so $\cos(\alpha/2) = r/d$, where α is the crank angle for the return stroke.

$\cos(\alpha/2) = 100/200 = 0.5$ → $\alpha/2 = 60°$ → **α = 120°** (return), cutting = $360° - 120° =$ **240°**

Ratio = 240/120 = **2.0**

Exam tip: The stroke length depends on the slotted-lever length, but the time ratio depends only on r/d.

Q7 **Answer: (C) $e^{-1/6}$**

Concept: A 1^{∞} form: take logs. $\ln L = \lim (1/x^2) \ln(\sin x/x)$.

$\sin x/x = 1 - x^2/6 + \ldots \rightarrow \ln(\sin x/x) \approx -x^2/6$

$\ln L = -1/6 \rightarrow \mathbf{L = e^{-1/6}}$

Exam tip: For 1^{∞} forms, $L = e^{\lim g(f-1)}$ when $f \rightarrow 1$ and $g \rightarrow \infty$. Here $(f-1)g = (-x^2/6)(1/x^2) = -1/6$.

Q8 **Answer: 0.098 (accept 0.097 to 0.099)**

Concept: Walli's formula: $\int_0^{\pi/2} \sin^m x \cos^n x \, dx = [(m-1)(m-3)\ldots][(n-1)(n-3)\ldots]/[(m+n)(m+n-2)\ldots] \times k$, where $k = \pi/2$ if m and n are both even, otherwise $k = 1$.

$m = 4, n = 2$: $(3 \times 1)(1)/(6 \times 4 \times 2) \times \pi/2 = (3/48)(\pi/2) = \mathbf{\pi/32}$

$\pi/32 = \mathbf{0.098}$

Exam tip: Only use the $\pi/2$ factor when both powers are even. For $\int \sin^3 x \cos^2 x$ the answer is $(2)(1)/(5 \times 3 \times 1) = 2/15$, with no π.

Q9 **Answer: (C) 4/3**

Concept: Find the intersections, then integrate (upper − lower).

$x^2 = 2x \rightarrow x = 0, 2$. On (0, 2) the line is above the parabola.

$A = \int_0^2 (2x - x^2) \, dx = 4 - 8/3 = \mathbf{4/3}$

Exam tip: Shortcut: the area between $y = x^2$ and any straight line cutting it at $x = a$ and $x = b$ is $(b - a)^3/6$. Here $(2 - 0)^3/6 = 4/3$.

Q10 **Answer: 208.3 (accept 206.3 to 210.3)**

Concept: The energy given out equals the drop in kinetic energy: $\Delta E = \frac{1}{2} I (\omega_1^2 - \omega_2^2)$, with $I = mk^2$.

$\omega_1 = 25.133$ rad/s, $\omega_2 = 22.619$ rad/s $\rightarrow \omega_1^2 - \omega_2^2 = \mathbf{120.01}$

$I = 2 \times 4500/120.01 = \mathbf{74.99\ kg \cdot m^2} \rightarrow m = 74.99/0.6^2 = \mathbf{208.3\ kg}$

Exam tip: $\Delta E = I\omega^2 C_s$ with the mean speed (228 rpm) gives the same answer, because $\frac{1}{2} I(\omega_1^2 - \omega_2^2) = I\omega_{mean}(\omega_1 - \omega_2)$ exactly when $\omega_{mean} = (\omega_1 + \omega_2)/2$.

Q11 **Answer: (A) $l_1 l_2 = k^2$**

Concept: Dynamic equivalence needs (i) the same total mass, (ii) the same CG position: $m_1 l_1 = m_2 l_2$, and (iii) the same moment of inertia: $m_1 l_1^2 + m_2 l_2^2 = mk^2$.

Combining the three gives $\mathbf{l_1 l_2 = k^2}$.

Exam tip: If one mass is placed at the small end, the other must be at the centre of percussion ($l_2 = k^2/l_1$), usually not at the big end. Placing them at both ends needs a correction couple.

Q12 **Answer: 1.56 (accept 1.54 to 1.57)**

Concept: $T = F\, r \sin(\theta + \beta)/\cos\beta$, where $\sin\beta = \sin\theta/n$.

$n = 5$: $\sin\beta = \sin 120°/5 = 0.1732 \rightarrow$ **$\beta = 9.97°$**

$T = 20\,000 \times 0.1 \times \sin 129.97° / \cos 9.97° = 1556$ N·m ≈ **1.56 kN·m**

Exam tip: At $\theta = 90°$ the expression gives exactly $F\, r\, (\sin(90° + \beta) = \cos\beta)$, a quick check. Past 90° the obliquity reduces the torque below $F\, r \sin\theta$ (1.73 kN·m here) because the rod pulls at an angle.

Q13 **Answer: (D) 160 rpm**

Concept: Train value = product of (driver teeth/driven teeth) for each stage.

$N_{out} = 1440 \times (20/60) \times (18/54) = 1440/9 =$ **160 rpm**, in the same direction as the input (two external meshes)

Exam tip: In a compound train each stage ratio multiplies, so a 9:1 reduction needs only two 3:1 stages instead of one huge gear.

Q14 **Answer: 100 (clockwise)**

Concept: Tabular method: arm fixed, sun $+x$ gives planet $-x(20/20)$ and annulus $-x(20/60)$. Then add the arm speed y to every member.

Clockwise positive: sun $y + x = 1000$; annulus $y - x/3 = -200$

Subtract: $(4/3)x = 1200 \rightarrow x = 900$, $y =$ **100 rpm clockwise**

Exam tip: Check the teeth: $20 + 2 \times 20 = 60$ ✓. Always write each member's speed as $y \pm$ (ratio)x and solve the two given conditions.

Q15 **Answer: (B) 4 times**

Concept: For every standard motion (SHM, UARM, cycloidal), $a_{max} \propto h\omega^2/\beta^2$, while $v_{max} \propto h\omega/\beta$.

Doubling ω multiplies the acceleration by $2^2 =$ **4** (and the velocity by 2).

Exam tip: Inertia force (and contact stress, spring force needed to keep contact) therefore grows with the square of speed. This is why high-speed cams need careful motion design.

What next?

Day 18 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 19/90: Theory of Machines, Governors, Balancing & Gyroscope. Comment **GATE90** on our reel to get all links.
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